

CARNARVON, Australia

WMO: 943000

Lat: 24.8878S

Lon: 113.6700E

Elev: 15

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.6	49.0	20.2	15.2	72.1	25.6	19.7	68.3	23.4	68.5	21.4	68.2	8.2	130	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.4	97.6	70.1	93.5	71.2	90.2	72.0	78.5	86.9	77.4	85.6	76.5	84.6	14.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	137.6	82.8	75.1	131.7	82.7	73.9	126.3	82.3	42.1	87.0	41.0	85.6	40.0	84.6	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	24.4	23.0	41.9	107.2	1.9	4.5	40.5	110.5	39.4	113.2	38.3	115.7	36.9	119.0		
(5)			36.8	80.9	2.1	3.1	35.3	83.1	34.1	84.9	32.9	86.6	31.3	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.0	80.0	81.7	80.5	75.6	69.3	64.1	62.0	63.5	66.4	70.2	74.0	77.1
	DBStd	7.76	4.15	4.25	4.37	4.15	4.04	3.66	3.53	3.28	3.47	3.33	3.76	3.92
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	268	0	0	0	0	8	60	106	69	24	1	0	0
	CDD50	8022	931	887	946	767	599	423	373	419	491	627	720	839
	CDD65	2814	466	467	481	317	142	33	14	23	64	163	270	374
	CDH74	22657	4286	4828	4795	2309	855	182	88	154	268	694	1496	2702
	CDH80	7410	1473	1871	1826	649	190	14	6	14	57	159	393	758
Wind	WSAvg	13.7	16.7	15.7	14.6	12.6	10.3	9.8	9.9	11.3	13.8	15.9	16.5	17.4
Precipitation	PrecAvg	8.4	0.4	0.8	0.6	0.6	1.2	1.8	1.9	0.7	0.2	0.2	0.1	0.0
	PrecMax	15.8	2.7	5.8	8.3	3.5	5.6	6.3	8.5	2.3	1.1	1.5	1.9	0.6
	PrecMin	3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.8	0.6	1.3	1.3	0.8	1.4	1.3	2.1	0.6	0.3	0.3	0.3	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.7	102.5	102.0	96.7	89.7	82.0	80.6	81.8	86.6	91.1	96.9	98.2
		MCWB	70.8	71.8	70.9	64.8	61.3	58.1	58.0	56.9	58.9	62.9	64.6	67.1
	2%	DB	94.4	95.9	95.8	90.1	85.0	78.6	76.4	78.0	80.0	84.0	89.0	90.9
		MCWB	73.8	74.1	70.4	67.0	62.3	59.1	59.2	58.9	61.0	63.8	66.0	71.1
	5%	DB	89.5	91.5	91.3	86.1	81.5	75.8	73.5	75.1	76.6	79.9	83.9	86.6
		MCWB	74.3	75.0	72.4	68.8	63.6	60.5	59.4	60.0	62.4	64.9	67.9	72.1
	10%	DB	86.6	88.6	88.2	83.2	78.5	73.3	71.2	72.5	74.1	77.2	80.4	83.7
		MCWB	74.0	75.3	72.7	69.7	64.5	60.2	59.5	60.0	62.1	65.0	68.2	71.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.7	80.5	79.0	76.4	73.2	70.2	69.1	68.2	69.8	70.9	73.3	78.3
		MCDB	87.4	88.8	87.9	82.8	78.3	73.4	73.6	72.7	75.3	78.6	81.4	83.9
	2%	WB	77.9	78.9	77.4	74.5	71.1	67.3	66.3	66.1	67.1	69.3	71.9	75.6
		MCDB	86.0	87.6	85.7	81.2	76.5	71.6	70.3	71.2	73.2	77.2	80.1	83.8
	5%	WB	76.7	77.9	76.4	73.2	69.7	65.6	63.9	64.4	65.5	68.1	70.8	74.0
		MCDB	85.1	86.3	84.5	80.4	76.3	70.4	68.6	70.1	72.3	75.6	79.0	82.4
	10%	WB	75.5	76.9	75.4	72.0	67.9	63.8	62.0	62.8	64.0	66.8	69.7	72.8
		MCDB	84.1	85.0	83.5	79.2	74.5	69.4	67.6	69.4	71.4	74.5	77.8	81.1
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.4	14.9	16.3	19.8	20.0	20.0	19.8	17.2	15.3	14.0	13.1
		MCDBR	19.5	20.0	22.0	22.4	25.3	24.3	23.8	25.4	22.7	22.1	22.0	18.5
		MCWBR	8.1	8.2	9.9	10.8	12.8	12.5	12.6	13.7	11.8	11.5	10.3	8.3
	5% WB	MCDBR	14.6	14.3	14.9	16.4	17.9	16.0	16.3	18.5	17.6	16.5	15.2	13.8
		MCWBR	6.9	6.5	7.2	8.4	9.9	9.8	10.7	11.3	9.9	8.9	7.5	6.7
Clear-Sky Solar Irradiance	taub	0.374	0.375	0.357	0.335	0.311	0.305	0.299	0.303	0.316	0.332	0.337	0.350	
	taud	2.511	2.534	2.567	2.604	2.638	2.636	2.646	2.609	2.545	2.510	2.519	2.529	
	Ebn at Noon	307	302	299	292	287	282	289	299	307	312	317	316	
	Edh at Noon	36	35	32	28	25	24	24	28	32	35	36	36	
All-Sky Solar Radiation	RadAvg	2587	2401	2078	1685	1375	1191	1266	1598	2016	2350	2592	2699	
	RadStd	105	91	121	83	82	54	89	60	47	76	73	103	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	-3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page